

Enhanced Medicinal Plant Leaf Classification via Transfer Learning: A VGG16-Driven Approach for Precise Feature Extraction

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Abstract—Medicinal plant identification is essential for holistic treatment systems, where accurate taxonomy ensures the safe and effective use of therapeutic properties. However, traditional methods often struggle with high feature variability and visual similarities among species, increasing the risk of misidentification. To address this, Herbplex, an Android application, employs the VGG16 deep learning model for transfer learning. VGG16's pretrained architecture extracts deep features, optimizing the learning process for accurate classification. The model was trained on a dataset of 35 medicinal plant species, achieving an impressive classification accuracy of 93%. Herbplex not only ensures precise taxonomy prediction but also provides detailed medicinal properties of identified plants, enhancing its utility in healthcare and agriculture. By facilitating reliable plant recognition, Herbplex supports advancements in natural medicine, botany, and therapeutic research. This innovative solution underscores the potential of AI in promoting sustainable plant based treatments, ensuring safety, and driving innovation in holistic healthcare systems.

Keywords—*medicinal plant classification, transfer learning, VGG16, deep learning, convolutional neural networks (CNNs)*

I. INTRODUCTION

Plants have been essential for human health for centuries, serving as the foundation for many traditional medicinal practices, including Ayurveda and indigenous medicine. The need for accurate identification of medicinal plants has grown, as misidentification can lead to improper treatment and health risks. However, distinguishing plant species is challenging due to the variability in their morphology and similarities between species, especially as traditional expertise wanes. Machine learning, particularly deep learning models like Convolutional Neural Networks (CNNs), offers a promising solution for plant identification by automating feature extraction from plant images and achieving high classification accuracy. Transfer learning with pretrained models such as VGG16 further supports this process, enhancing performance even with limited training data. Despite progress, challenges persist in practical applications. Data limitations and the generalizability of models are key concerns. While transfer learning can produce high accuracy with small datasets,

adapting to diverse and larger datasets, especially on variable hardware, remains difficult.

Data augmentation can help improve accuracy but cannot fully prevent overfitting, particularly with limited data variation. Additionally, real time application and usability in mobile environments continue to be areas that need more attention. This research aims to create an efficient, real time solution for medicinal plant identification suitable for mobile devices, contributing to biodiversity monitoring, ecological research, and natural medicine. The objective is to develop a high-accuracy model using VGG16 and optimize its performance through transfer learning. The contributions of this study include: a robust model utilizing VGG16 transfer learning achieving 92% classification accuracy, demonstrating the effective application of VGG16 for mobile based solutions while managing computational limitations, and advancing mobile-based deep learning to bridge the gap between research and practical use in medicinal plant identification. The paper is organized as follows: Section 2 reviews related literature, discussing previous approaches and their strengths and limitations. Section 3 outlines the methodology, including the dataset, model architecture, and training process. Section 4 presents experimental results and analysis. Section 5 concludes with key findings and future research directions.

II. RELATED WORK

Ayumi et. al.[1] leveraged the VGG16 model in a transfer learning framework to identify medicinal plants, demonstrating high accuracy on small datasets but lacking evaluation for scalability and robustness on diverse datasets. Kanda et. al.[2] combined GANs, CNNs, and logistic regression for plant classification, achieving improved accuracy through data augmentation but facing challenges with computational demands and adaptability to new species. Sachar et. al.[3] proposed an ensemble approach using MobileNetV2, InceptionV3, and ResNet50, attaining 99.66% accuracy, though practical implementation may be constrained by computational requirements and generalizability issues.

The DIMPSAR dataset by Pushpa B R and N. Shobha Rani[4] provides 12,800 images spanning 40 and 80 plant species captured under diverse conditions in Karnataka and Kerala, aiding in medicinal plant research and machine learning development. Despite its breadth, it falls short of representing all medicinal plants, limiting its generalizability across regions and species. Arunaggi et al.[5] leveraged deep learning with transfer learning and PCA for improved plant classification accuracy but faced challenges in scalability and handling diverse datasets.

Ghosh et al.[11] introduce a hybrid transfer learning technique for the identification of medicinal plants. The study combines pre-trained deep learning models with domain-specific fine-tuning to enhance classification accuracy. By leveraging transfer learning, the approach significantly reduces the need for large labeled datasets, making it more practical for identifying medicinal plants in real-world applications where data might be limited or difficult to obtain. Dammavalam et al.[10] demonstrated transfer learning's effectiveness for leaf classification but struggled with imbalanced datasets and computational demands, impacting real-time feasibility. Putri et al.[12] propose a hybrid transfer learning technique for the identification of medicinal plants, showcasing its effectiveness in improving classification accuracy using deep learning models in the context of plant recognition.

Kaur and Kaur[7] on machine learning for plant species, Wong et al.[13] adapting transfer learning for mobile disease detection, and Zhao et al.[15] introducing a dataset for medicinal plant classification, although reliance on large labeled data constrained scalability. Sharmila et. al.[8], explores techniques for enhancing satellite image classification accuracy by focusing on image denoising and resolution enhancement. The study helps to improve the quality of data used for classification tasks in remote sensing.

Mimi et. al.[9], explore the use of computer vision techniques for detecting plant diseases, highlighting the potential of automated systems in agricultural diagnostics. They discuss various image processing and machine learning approaches to accurately identify symptoms of plant diseases, aiming to improve early detection and crop management. Patel, Gandhi, and Shah[14], propose an automatic medicinal plant identification system using Convolutional Neural Networks (CNNs). The study emphasizes the effectiveness of CNNs in extracting relevant features from plant images, enabling accurate classification of medicinal plants.

These efforts collectively advance plant identification research, emphasizing the need for scalable, efficient, and diverse solutions.

III. PROPOSED METHOD

This study introduces an optimized approach using a VGG16 transfer learning model for the classification of medicinal plant species. The proposed methodology leverages pretrained network features, finetuned to adapt to a dataset of plant images. Through preprocessing steps including image resizing, grayscale conversion, and data augmentation, the model is trained to improve its generalization capabilities. The use of transfer learning allows for efficient feature extraction and robust performance in plant species classification, validated by standard evaluation metrics like accuracy, precision, recall, and F1-score.

A. Data Collection

The dataset captures a wide variety of leaf images under different lighting and environmental conditions, ensuring that the model can be effectively generalized to different scenarios. To further strengthen the robustness of the classification model, the dataset was supplemented with additional images from publicly available botanical image repositories. These additional images increase the diversity of the dataset. Fig.1 illustrates sample images from the dataset used to train the model [6].

B. Data Preprocessing

All images were resized to 256x256 pixels for consistency, and pixel values were normalized to [0, 1] to improve training efficiency. Data augmentation techniques like random rotations, shifts, shear, zooming, and horizontal flipping were applied to enhance generalization and reduce overfitting. These augmentations expanded the dataset, helping the model learn invariant features and improving its robustness in medicinal plant classification.

C. Transfer Learning

Transfer learning is a technique that allows a model to apply knowledge from a task it has already learned to enhance its performance on a new related task. This is particularly useful in areas such as computer vision, where large labeled datasets are often difficult to obtain. In this study, transfer learning is applied using a pre-trained VGG16 model, which is known to achieve high accuracy in image classification tasks. By reusing the feature extraction features of VGG16, a custom classification layer is added specifically for medicinal plant species identification based on leaf images, reducing the need for extensive training data and time.

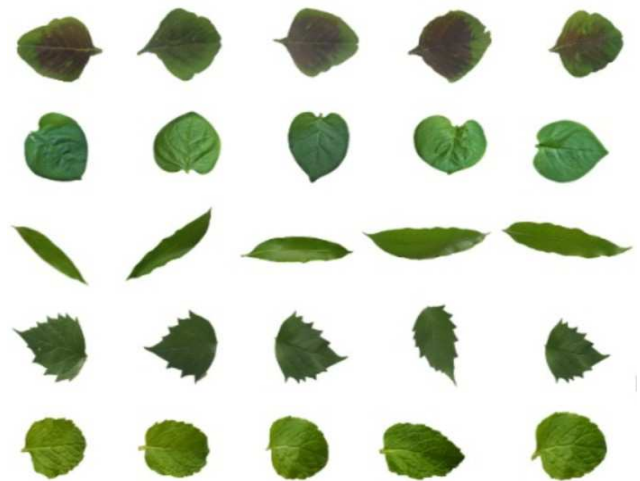


Fig. 1. Image samples of Medicinal plants

As shown in the Fig.2 VGG16 is first trained on a general dataset (Task 1) to learn transferable features. For the medicinal plant classification task (Task 2), the prebuilt VGG16 layers are retained, while a custom output layer is added and finetuned to classify different plant species. This approach increases the accuracy and efficiency of the model, allowing it to perform well even with a limited data set, proving the robustness of VGG16 as a feature extractor in specialized classification tasks.

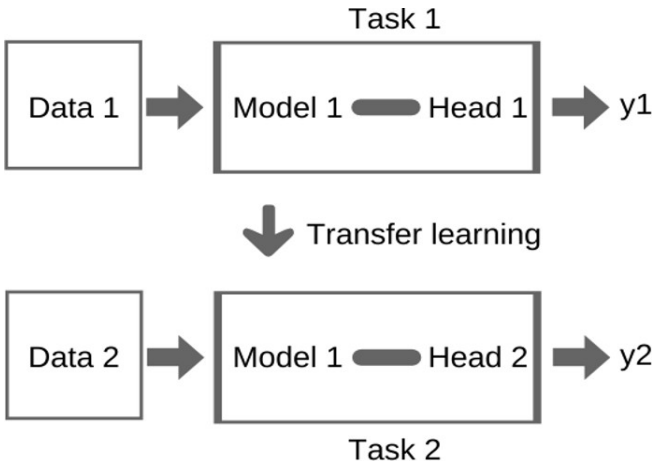


Fig. 2. Transfer Learning Workflow.

D. VGG16 Deep Learning Architecture

VGG16, a CNN architecture known for its strong performance in image classification, is particularly effective in identifying medicinal plants. It processes 224×224 RGB images through 16 layers (13 convolutional and 3 fully connected). The use of 3×3 filters helps capture detailed spatial hierarchies, while ReLU activation adds nonlinearity for learning complex features. The final fully connected layer of VGG16 uses a softmax function for multi-class classification. Its ability to fine-tune pretrained weights boosts accuracy, even with small datasets like medicinal plant identification. This adaptability makes VGG16 a valuable tool for improving plant classification, supporting biodiversity conservation, and the sustainable use of medicinal plants. Fig.3 provides a more structured view of the VGG16 architecture, showing each layer's input and output dimensions. The architecture starts with an input of 224×224 , which goes through multiple convolutional layers with ReLU activation and max pooling, reducing the input down to a 7×7 feature map before reaching the fully connected layers. The final fully connected layer uses a softmax function for multiclass classification, enabling the identification of plant species from the leaf images. Finetuning the pretrained weights from VGG16 enhances the model's accuracy, even when working with smaller datasets like medicinal plant identification. The finetuned VGG16 model was tested on a validation dataset to evaluate its performance. The adaptability of VGG16, particularly in tasks with limited data, makes it a valuable tool for applications like plant classification, where biodiversity conservation and the sustainable use of medicinal plants are of great importance. Despite the emergence of newer architectures, VGG16's simplicity and strong feature extraction capabilities continue to make it a reliable choice for a wide range of image classification tasks.

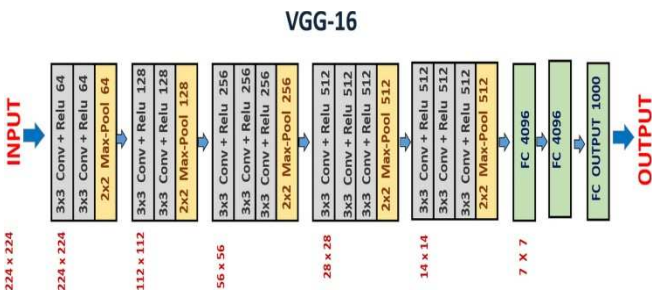


Fig. 3. VGG16 Architecture Configuration

The pseudocode explains how to train the VGG16 model step by step. It starts by setting up the model with initial values for its weights. Then, the training process runs for several rounds, called epochs. In each round, the data is shuffled to mix it up and is split into smaller groups, called mini-batches, to make it easier to process.

For each mini-batch, the model makes predictions based on its current settings. These predictions are checked against the correct answers to see how far off they are, which is called the loss. The model then works backward to figure out how to adjust its settings to improve its accuracy. Finally, it updates its settings slightly to reduce the loss. This process is repeated for all the mini-batches and epochs, gradually improving the model's ability to make better predictions.

Algorithm: Training a Model using VGG16 model

Require: Dataset D , Learning rate α , Batch size B , Number of epochs E

- 1: Initialize model weights W
- 2: for epoch = 1, 2, ..., E do
 - 3: Shuffle dataset D
 - 4: Divide D into mini-batches of size B
 - 5: for each batch b in mini-batches do
 - 6: Perform a forward pass:
 - Compute predictions P using weights W
 - 7: Calculate the loss:
 - $L_{batch} = L(P, \text{GroundTruth})$
 - 8: Perform a backward pass:
 - Compute gradients ∇W of L_{batch} with respect to model weights W
 - 9: Update weights:
 - $W = W - \alpha \nabla W$
 - 10: end for
- 11: end for

IV. RESULTS AND DISCUSSIONS

The experiments conducted for this study were carried out on a Windows OS system equipped with an Intel® Core™ i7 10700K CPU, 16 GB RAM, and an NVIDIA RTX 3070 GPU, providing the robust computational power required for handling intensive deep learning tasks. This setup ensured efficient training, evaluation, and finetuning of models, enabling faster processing of large datasets and optimization of parameters. In this section, we present a comprehensive analysis of the experimental results, comparing various performance metrics such as accuracy, precision, recall, and F1- scores across different deep learning models. The results emphasize the effectiveness of transfer learning in leveraging pretrained architectures to classify leaf images accurately, even with limited domain specific data. Furthermore, the discussion highlights the superior performance of ensemble approaches, which combine predictions from multiple models to achieve higher classification accuracy and reliability. These findings underline the potential of such techniques in advancing the automation of medicinal plant identification.

A. VGG16 Model Evaluation

TABLE I. ACCURACY RESULT OF VGG16 TRANSFER LEARNING MODEL FOR MEDICINAL PLANT CLASSIFICATION

No of Epochs	Batch size	Learning rate	Test Accuracy	Loss function
10	32	0.01	92.57	0.247
10	32	0.01	92.69	0.217
10	32	0.01	92.42	0.3097

The TABLE I illustrates the performance of a transfer learning model used for classifying medicinal plants, with various training configurations involving the number of epochs, batch size, and learning rate. By comparing these configurations, we can observe the model's accuracy and loss function values, which are crucial for evaluating its effectiveness in recognizing different plant species based on leaf characteristics.

The model was trained and evaluated across three distinct epochs (10, 25, and 50), with a fixed batch size of 32 and a learning rate of 0.01. Notably, the model achieved its highest test accuracy of 92.69% and the lowest loss function value of 0.217 at 25 epochs, indicating that this configuration strikes an optimal balance between model generalization and error minimization. Increasing the number of epochs to 50 did not improve the model's performance, as the test accuracy slightly decreased to 92.42%, while the loss function value increased to 0.3097, suggesting overfitting. Conversely, training the model for only 10 epochs resulted in a slightly lower test accuracy of 92.57% and higher loss of 0.247 compared to 25 epochs.

Overall, the analysis reveals that 25 epochs with the chosen hyperparameters (learning rate and batch size) yield the most balanced and accurate results, effectively capturing the distinguishing features of medicinal plants without overfitting, making this configuration ideal for the classification task. The classification report presented in Fig. 4 illustrates the performance metrics for the medicinal plant species identification model built using the VGG16 architecture with transfer learning. This detailed report includes key evaluation metrics such as precision, recall, F1-score, and support for each plant species, encompassing a range of examples like Aloe Barbadensis Miller (Aloe Vera), Alpinia Galanga (Rasna), and Amaranthus Viridis (Arive-Dantu), among others. The model demonstrates strong performance across the majority of species, achieving an impressive overall accuracy of 93%. Additionally, the macro and weighted average precision scores, recall, and F1-scores are consistently high, hovering around 93% and 94%, respectively. These results highlight the model's capability to reliably identify different medicinal plant species, underscoring its potential for effective use in real world applications for plant classification and identification tasks. Fig.5 shows the training and validation of the VGG16 model with PCA applied for plant species classification. The left graph in Fig.5 illustrates the training and validation accuracies over 10 epochs. Both training and validation accuracies increase rapidly in the initial epochs and reach convergence around the 4th epoch. Up to the last epoch, the model achieves nearly 95% accuracy, indicating strong performance and efficient generalization on the validation set. The right graph in Fig.5 shows the training and validation loss, with both curves falling sharply during the initial epochs, indicating a rapid reduction in errors. The training and validation losses stabilize and converge by the 4th epoch and maintain low values for the rest of the training. This steady convergence of accuracy and loss indicates that the model learned efficiently with minimal overfitting.

B. Comparison with other Models

In our study, we found that the VGG16 model outperformed other deep learning architectures for medicinal plant classification using leaf images, achieving an impressive 92.69% accuracy along with strong Precision, Recall, and F1

Score metrics of 0.92, indicating consistent and reliable performance. This result aligns with the state-of-the-art models in the field, where VGG16 has been frequently used for image classification tasks due to its effective feature extraction capabilities through its 16-layer architecture. Our analysis showed that ResNet50, known for its skip connections that help improve gradient flow and model depth, performed well with an accuracy of 87.54 percent and an F1 Score of 0.90 but demanded higher computational resources, limiting its application in resource-constrained scenarios. DenseNet121, with its feature reuse strategy promoting efficient learning, demonstrated comparable results with an 88.53% accuracy and an F1 Score of 0.88, showing strong performance but also requiring substantial memory, which may not be ideal for mobile or embedded system deployment. MobileNetV2, designed for speed and efficiency with depth wise separable convolutions, achieved 88.67% accuracy and an F1 Score of 0.86, proving suitable for real time applications where computational efficiency is crucial, even if slightly less accurate than VGG16. The PCA based VGG16, while intended to simplify calculations through dimensionality reduction, performed poorly with an accuracy of only 73.67% and reduced scores in Precision, Recall, and F1, suggesting that essential feature information may have been lost during the process, ultimately hindering its classification capability. Table II shows the evaluation metrics of each model. Our findings echo existing literature that supports VGG16's use in plant classification due to its balanced trade-off between performance and resource consumption, and contrast with other studies that show while models like DenseNet121 and ResNet50 can achieve higher accuracy, they come at a significant computational cost. MobileNetV2's performance confirms its suitability for applications that prioritize efficiency, but with a slight sacrifice in accuracy. This comparative analysis underscores VGG16's robustness for medicinal plant classification tasks, offering a high level of accuracy without the demanding computational requirements of deeper architectures. Future work should consider expanding the dataset with more varied images, applying advanced augmentation strategies, and combining different models in a hybrid approach to push the performance boundaries of classification models in botanical research, especially for real-time and field-based applications.

	precision	recall	f1-score	support
Aloe Barbadensis Miller (Aloe Vera)	0.88	1.00	0.94	23
Alpinia Galanga (Rasna)	0.83	1.00	0.91	10
Amaranthus Viridis (Arive-Dantu)	1.00	0.96	0.98	24
Artocarpus Heterophyllus (Jackfruit)	1.00	1.00	1.00	11
Azadirachta Indica (Neem)	0.86	1.00	0.92	12
Basella Alba (Basella)	0.93	0.70	0.80	20
Brassica Juncea (Indian Mustard)	1.00	0.50	0.67	6
Carica papaya (Papaya)	1.00	1.00	1.00	27
Carissa Carandas (Karanda)	1.00	1.00	1.00	14
Catharanthus roseus (Periwinkle)	1.00	1.00	1.00	8
Citrus Limon (Lemon)	0.92	1.00	0.96	11
Curcuma longa (Turmeric)	1.00	1.00	1.00	7
Ficus Auriculata (Roxburgh fig)	0.64	0.90	0.75	10
Ficus Religiosa (Peepal Tree)	1.00	1.00	1.00	12
Hibiscus Rosa-sinensis	1.00	0.88	0.93	8
Jasminum (Jasmine)	0.92	0.79	0.85	14
Mangifera Indica (Mango)	1.00	0.67	0.80	12
Mentha (Mint)	0.95	1.00	0.97	19
Moringa Oleifera (Drumstick)	0.88	1.00	0.94	15
Muntingia Calabura (Jamaica Cherry-Gaagase)	0.85	1.00	0.92	11
Murraya Koenigii (Curry)	1.00	0.67	0.80	12
Musa paradisiaca Linn (Banana)	1.00	1.00	1.00	31
Nerium Oleander (Oleander)	1.00	1.00	1.00	12
Nyctanthes Arbor-tristis (Parijata)	1.00	1.00	1.00	8
Ocimum Tenuiflorum (Tulsi)	1.00	0.80	0.89	10
Piper Betle (Betel)	0.90	1.00	0.95	9
Plectranthus Amboinicus (Mexican Mint)	0.90	1.00	0.95	9
Pongamia Pinnata (Indian Beech)	1.00	1.00	1.00	12
Psidium Guajava (Guava)	1.00	1.00	1.00	13
Punica Granatum (Pomegranate)	0.78	0.93	0.85	15
Santalum Album (Sandalwood)	0.79	1.00	0.88	11
Syzygium Cumini (Jamun)	1.00	0.86	0.92	7
Syzygium Jamboe (Rose Apple)	1.00	0.82	0.90	11
Tabernaemontana Divaricata (Crape Jasmine)	0.73	0.73	0.73	11
Trigonella Foeniculum-graecum (Fenugreek)	0.86	0.86	0.86	7
accuracy			0.93	452
macro avg	0.93	0.92	0.92	452
weighted avg	0.94	0.93	0.93	452

Fig. 4. Classification report of the model over 25 epochs

Overall, VGG16 emerged as the top performer, achieving a strong balance between high accuracy and computational efficiency for medicinal plant classification using leaf images. Its straightforward architecture, comprising 16 layers, proved effective in capturing essential features while maintaining reliable performance metrics, which set it apart from the other models tested. In contrast, the PCA-based VGG16 variant suffered a significant performance drop due to the potential loss of critical features during dimensionality reduction, which hindered its ability to effectively classify the plant species. On the other hand, ResNet50 and DenseNet121, with their deeper and more complex architectures designed to improve feature learning through mechanisms like skip connections and feature reuse, demonstrated robust

performance. However, these models come with higher computational and memory demands, making them less ideal for scenarios where limited resources are available. MobileNetV2, while slightly less accurate with an F1 Score of 0.86, proved to be an efficient model optimized for fast inference and minimal computational cost, making it suitable for deployment in real-time, resource-constrained environments. This comparative analysis underscores the nuanced trade-offs between model accuracy, computational efficiency, and resource requirements, highlighting VGG16's effectiveness as an optimal choice for balancing these factors. It also sheds light on the suitability of MobileNetV2 for real-world applications where quick decision-making is critical, despite a slight trade-off in overall accuracy.

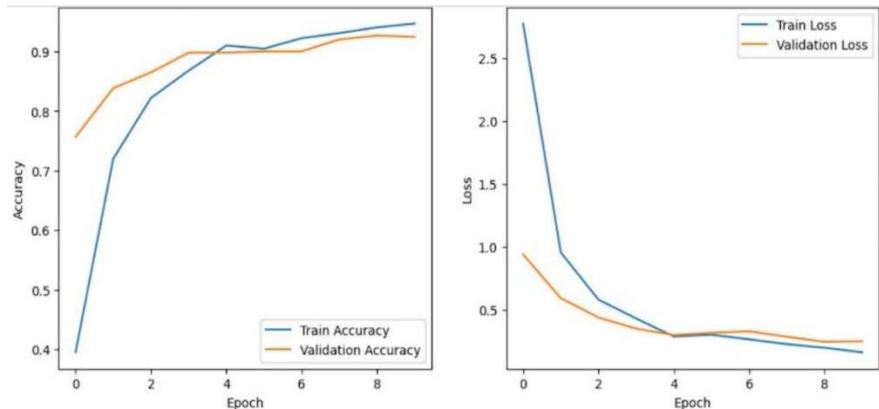


Fig. 5. Accuracy chart and loss chart for 10 epochs

TABLE II. MODEL COMPARISON WITH OTHER STATE-OF-THE-ART TRANSFER LEARNING MODELS

Models	Accuracy	Precision	Recall	F1 Score
VGG16	92.69	0.93	0.92	0.92
ResNet	87.54	0.903	0.85	0.87
MobileNet V2	88.67	0.88	0.86	0.86
DenseNet 121	88.53	0.89	0.87	0.88
PCA based VGG16	73.67	0.80	0.75	0.77

V. CONCLUSION

In this research, we proposed an advanced transfer learning model for the classification of medicinal plants using leaf identification. This model was designed to leverage the power of pre-trained deep learning models, enabling accurate and efficient classification. We evaluated six different pre-trained models to identify the best-performing one for this specific task: VGG16, ResNet50, DenseNet121, PCA-based VGG16, and MobileNetV2. Each of these models was assessed based on their performance in classifying medicinal plants from images of their leaves, with a focus on achieving high accuracy while maintaining computational efficiency. Our chosen model incorporated a fully connected layer with 200 neurons, a learning rate of 0.001, and a batch size of 32. After training and fine-tuning, our model achieved an impressive accuracy of approximately 93%, outperforming all other models tested. This result highlights the effectiveness of transfer learning, particularly in the domain of plant classification, and underscores the potential of deep learning techniques to revolutionize botanical research, especially in automated plant identification for medicinal purposes. Despite the promising results, we also encountered challenges associated with the high memory consumption required by these deep learning models. The models, while offering superior accuracy, demand significant computational

resources, making them less suitable for deployment in embedded systems or mobile devices with limited processing power and memory. This presents a limitation for real-world applications, particularly in resource-constrained environments such as fieldwork or remote areas. The research on "Enhanced Medicinal Plant Leaf Classification via Transfer Learning: A VGG16-Driven Approach for Precise Feature Extraction" establishes a solid foundation for future advancements with numerous opportunities for growth. Expanding the dataset to include rare and region-specific plant species would enhance its global applicability. Integrating offline functionality could benefit users in remote areas with limited internet access, while adding machine learning models for disease detection could provide valuable insights into plant health and treatment. Encouraging user contributions to enrich the dataset and exploring augmented reality (AR) for interactive visualization would further enhance user engagement. Multilingual support could broaden accessibility, and integration with external databases or APIs could provide deeper insights into plant care and traditional uses. Collaborating with botanical gardens and research institutions would ensure scientific accuracy and credibility. These enhancements would transform the research into a comprehensive platform for botanical knowledge and sustainable practices.

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